



第一章 作业

北京理工大学

1、化简下列复数，并求出它的实部、虚部、共轭复数、模和辐角主值

$$(3) \ i^{10} - 6i^{15} + i$$

$$\text{解: } i^{10} - 6i^{15} + i \quad \text{实部: } -1$$

$$\text{虚部: } 7$$

$$= i^2 - 6i^3 + i$$

$$\text{共轭复数: } -1 - 7i$$

$$= -1 + 7i$$

$$\text{模: } \sqrt{(-1)^2 + 7^2} = 5\sqrt{2}$$

$$\text{辐角主值: } \arg Z = \arctan(-7) + \pi = \pi - \arctan 7$$

5、将下列复数化成三角表示式和指数表示式

$$(6) \frac{(\cos\varphi - i\sin\varphi)^3}{(\cos 2\varphi + i\sin 2\varphi)^2}$$

解：
$$\frac{(\cos\varphi - i\sin\varphi)^3}{(\cos 2\varphi + i\sin 2\varphi)^2} = \frac{[\cos(-\varphi) + i\sin(-\varphi)]^3}{(\cos 2\varphi + i\sin 2\varphi)^2}$$

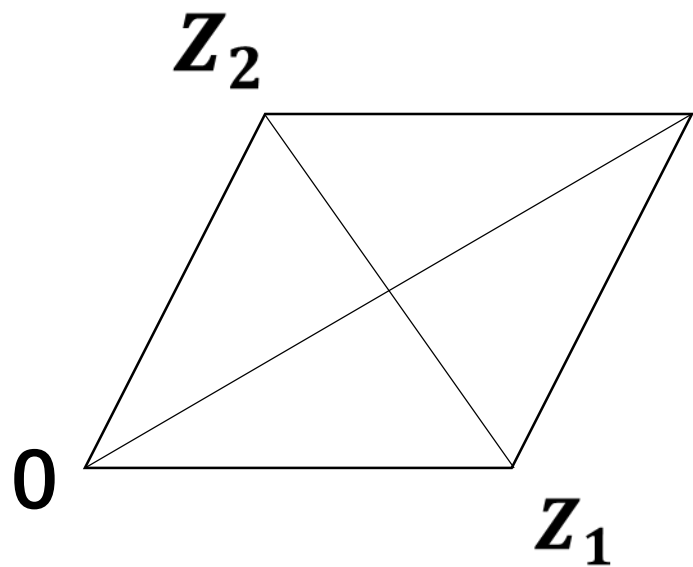
$$= \frac{(e^{-i\varphi})^3}{(e^{2i\varphi})^2} = \frac{e^{-3i\varphi}}{e^{4i\varphi}} = e^{-7i\varphi}$$

$$= \cos(-7\varphi) + i\sin(-7\varphi)$$

6、证明： $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$ ，并说明其几何意义。

$$\begin{aligned} \text{证明：} & |z_1 + z_2|^2 + |z_1 - z_2|^2 \\ &= (z_1 + z_2)\overline{(z_1 + z_2)} + (z_1 - z_2)\overline{(z_1 - z_2)} \\ &= (z_1 + z_2)(\overline{z_1} + \overline{z_2}) + (z_1 - z_2)(\overline{z_1} - \overline{z_2}) \\ &= z_1\overline{z_1} + z_1\overline{z_2} + z_2\overline{z_1} + z_2\overline{z_2} + z_1\overline{z_1} - z_1\overline{z_2} - z_2\overline{z_1} + z_2\overline{z_2} \\ &= 2z_1\overline{z_1} + 2z_2\overline{z_2} \\ &= 2(|z_1|^2 + |z_2|^2) \end{aligned}$$

几何意义：平行四边形对角线平方和等于边长的平方和



8、求下列各式的值

$$(2) (-1+i)^4; \quad (3) \sqrt[6]{1};$$

解: (2) $(-1+i)^4$

$$= \sqrt{2}^4 \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)^4 = \sqrt{2}^4 (\cos 3\pi + i \sin 3\pi) = -\sqrt{2}^4 = -4$$

(3) $\sqrt[6]{1}$

$$= (\cos 0^\circ + i \sin 0^\circ)^{\frac{1}{6}} = \cos \frac{2k\pi}{6} + i \sin \frac{2k\pi}{6}, \quad k = 0, 1, 2, 3, 4, 5$$

$$= \left\{ 1, \frac{1}{2} + \frac{\sqrt{3}}{2}i, -\frac{1}{2} + \frac{\sqrt{3}}{2}i, -1, -\frac{1}{2} - \frac{\sqrt{3}}{2}i, \frac{1}{2} - \frac{\sqrt{3}}{2}i \right\}$$

10、指出下列各题中点Z的轨迹或所在范围，并作图

$$(5) |z+3| + |z+1| = 4;$$

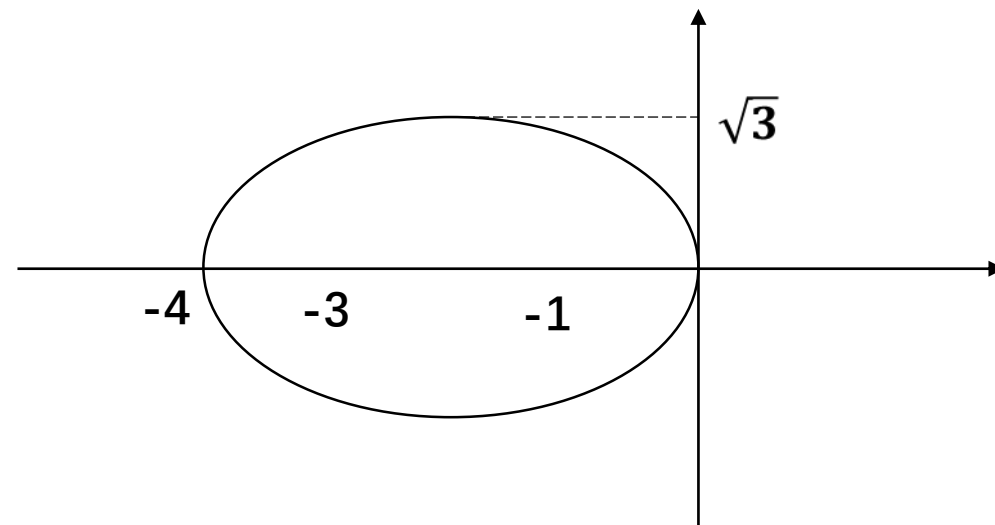
解： $|z+3| + |z+1| = 4;$

$$4 = 2a \longrightarrow a = 2$$

$$|-3 - (-1)| = 2c$$

$$\longrightarrow c = 1$$

$$\therefore b = \sqrt{a^2 - c^2} = \sqrt{3}$$



$\therefore$ 是长轴在X轴且长度为4，短轴
平行于 y轴且长度为 $2\sqrt{3}$ 的椭圆

11、用复参数方程表示下列各曲线

(3) 椭圆 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$;

解：

令 $\begin{cases} x=acost \\ y=bsint \end{cases}$

则满足所给椭圆方程

$$\therefore z=acost+ibsint, t \in [0, 2\pi]$$

12. 函数 $w = \frac{1}{z}$ 把下列 z 平面上的曲线映射成 w 平面上怎样的曲线?

(3) $x = 2$;

$$\text{解: } w = \frac{1}{z} = \frac{2}{4+y^2} - \frac{y}{4+y^2}i \Rightarrow \begin{cases} u = \frac{2}{4+y^2} \\ v = -\frac{y}{4+y^2} \end{cases}$$

$$\Rightarrow u^2 + v^2 = \frac{u}{2} \Rightarrow \left(u - \frac{1}{4}\right)^2 + v^2 = \frac{1}{16}$$

$\therefore$ 映射到 w 平面后是以 $\left(\frac{1}{4}, 0\right)$ 为圆心, 半径为 $\frac{1}{4}$ 的圆.

14、证明： $\lim_{z \rightarrow 0} \frac{\operatorname{Re}(z)}{z}$ 不存在

证明：令 $z=x+iy$ ，则 $\frac{\operatorname{Re}(z)}{z} = \frac{x}{x+iy}$

$$\text{令 } y=kx, \text{ 则 } \lim_{z \rightarrow 0} \frac{\operatorname{Re}(z)}{z} = \lim_{x \rightarrow 0} \frac{x}{x+ikx} = \frac{1}{1+ik}$$

当 k 不同时极限值不同，所以原极限不存在



第二章 作业

北京理工大学

1. 利用导数定义求下列导数

(2) $f(z)=z\operatorname{Re}z$

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{(z + \Delta z) \operatorname{Re}(z + \Delta z) - z \operatorname{Re} z}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{\Delta z \operatorname{Re}(z + \Delta z) + z \operatorname{Re} \Delta z}{\Delta z}$$

$$z = 0 \text{ 时, } f'(z) = \lim_{\Delta z \rightarrow 0} \frac{\Delta z \operatorname{Re} \Delta z}{\Delta z} = 0 \Rightarrow f'(z)|_{z=0} = 0$$

$$\begin{aligned} z \neq 0 \text{ 时, } \lim_{\Delta z \rightarrow 0} (\operatorname{Re}(z + \Delta z) + z \frac{\Delta x}{\Delta x + i\Delta y}) &= \operatorname{Re}(z) + \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} z \frac{\Delta x}{\Delta x + i\Delta y} \\ &= \operatorname{Re}(z) + \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y = k\Delta x}} z \frac{\Delta x}{\Delta x + ik\Delta x} = \operatorname{Re}(z) + z \frac{1}{1 + ik} \end{aligned}$$

当 k 取不同值时, 上述极限为不同值,
所以 $z \neq 0$ 时, $f(z)$ 不可导.

3、判断下列函数在何处可导，并求出其导数

$$(2) f(z) = \frac{2z}{z^2 + 1}$$

解：有理分式函数在分母不为0的区域内解析，

$$z^2 = -1 \Rightarrow z = \pm i$$

是奇点，不可导，在其他点处均可导且

$$f'(z) = \left(\frac{2z}{z^2 + 1} \right)' = \frac{2(z^2 + 1) - 2z(2z)}{(z^2 + 1)^2} = \frac{2(1 - z^2)}{(z^2 + 1)^2}$$

5、试证下列函数在复平面上解析，并分别求出其导数

$$(1) f(z) = x^3 + 3x^2 yi - 3xy^2 - y^3 i$$

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2 \quad \frac{\partial v}{\partial y} = 3x^2 - 3y^2$$

$$\frac{\partial u}{\partial y} = -6xy \quad \frac{\partial v}{\partial x} = 6xy$$

四个偏导均连续，且满足C-R方程

$\therefore$ 在复平面上处处解析

$$\begin{aligned} f'(z) &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = 3x^2 - 3y^2 + i6xy \\ &= 3(x^2 - y^2 + i2xy) = 3z^2 \end{aligned}$$

7、 设 $f(z) = \begin{cases} \frac{x^3 - y^3 + i(x^3 + y^3)}{x^2 + y^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$

证明： (1) $f(z)$ 在 $z=0$ 连续；

(2) $f(z)$ 在 $z=0$ 满足C-R方程；

(3) $f(z)$ 在 $z=0$ 不可导。

证明: (1) $f(z)$ 在 $z=0$ 连续;

方法1: $\because x \rightarrow 0, y \rightarrow 0$ 时,

$(x^3 - y^3)$ 和 $(x^3 + y^3)$ 都是 $(x^2 + y^2)$ 的高阶无穷小,

$$\therefore \lim_{z \rightarrow 0} f(z) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^3 - y^3 + i(x^3 + y^3)}{x^2 + y^2} = 0 = f(0)$$

$\therefore f(z)$ 在 $z=0$ 连续.

方法2: 令 $x = r \cos \theta, y = r \sin \theta$, 则

$$\begin{aligned} \lim_{z \rightarrow 0} f(z) &= \lim_{r \rightarrow 0} \frac{r^3 (\cos^3 \theta - \sin^3 \theta) + ir^3 (\cos^3 \theta + \sin^3 \theta)}{r^2} \\ &= \lim_{r \rightarrow 0} r (\cos^3 \theta - \sin^3 \theta) + ir (\cos^3 \theta + \sin^3 \theta) \\ &= 0 \quad (\because \cos^3 \theta - \sin^3 \theta \text{ 和 } \cos^3 \theta + \sin^3 \theta \text{ 有界}) \\ &= f(0) \Rightarrow f(z) \text{ 在 } z=0 \text{ 处连续.} \end{aligned}$$

方法3:

$$\begin{aligned}|f(z) - f(0)| &= \left| \frac{x^3 - y^3 + i(x^3 + y^3)}{x^2 + y^2} \right| = \frac{|x^3 + iy^3 + i(x^3 + iy^3)|}{|z|^2} \\ &\leq \frac{|x^3 + iy^3| + |i(x^3 + iy^3)|}{|z|^2} = \frac{2\sqrt{x^6 + y^6}}{|z|^2}\end{aligned}$$

$$\because |z|^3 = \left(\sqrt{x^2 + y^2} \right)^3 = \sqrt{x^6 + y^6 + 3x^4y^2 + 3x^2y^4} \geq \sqrt{x^6 + y^6}$$

$$\therefore |f(z) - f(0)| \leq \frac{2\sqrt{x^6 + y^6}}{|z|^2} \leq \frac{2|z|^3}{|z|^2} = 2|z|$$

对 $\forall \varepsilon > 0$, $\exists \delta = \frac{\varepsilon}{2} > 0$, 使得当 $0 < |z - 0| < \delta$ (即 $0 < |z| < \frac{\varepsilon}{2}$ 时)

$$|f(z) - f(0)| \leq 2|z| < \varepsilon$$

$\therefore \lim_{z \rightarrow 0} f(z) = f(0) = 0 \Rightarrow f(z)$ 在 $z = 0$ 连续

证明：(2) $f(z)$ 在 $z=0$ 满足C-R方程；

$$u(x, y) = \frac{x^3 - y^3}{x^2 + y^2}, \quad v(x, y) = \frac{x^3 + y^3}{x^2 + y^2}$$

$$\left. \frac{\partial u}{\partial x} \right|_{z=0} = \lim_{\substack{\Delta x \rightarrow 0 \\ y=0}} \frac{u(\Delta x, y) - u(0, y)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\frac{\Delta x^3}{\Delta x^2}}{\Delta x} = 1, \quad \left. \frac{\partial v}{\partial y} \right|_{z=0} = \lim_{\substack{\Delta y \rightarrow 0 \\ x=0}} \frac{v(x, \Delta y) - v(x, 0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{\frac{\Delta y^3}{\Delta y^2}}{\Delta y} = 1,$$

$$\left. \frac{\partial u}{\partial y} \right|_{z=0} = \lim_{\substack{\Delta y \rightarrow 0 \\ x=0}} \frac{u(x, \Delta y) - u(x, 0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{\frac{-\Delta y^3}{\Delta y^2}}{\Delta y} = -1,$$

$$\left. \frac{\partial v}{\partial x} \right|_{z=0} = \lim_{\substack{\Delta x \rightarrow 0 \\ y=0}} \frac{v(\Delta x, y) - v(0, y)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\frac{\Delta x^3}{\Delta x^2}}{\Delta x} = 1,$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 1, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = -1$$

$\therefore$ 满足C-R方程.

证明：(3) $f(z)$ 在 $z=0$ 不可导

$$\begin{aligned} f'(0) &= \lim_{\Delta z \rightarrow 0} \frac{f(\Delta z) - f(0)}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{f(\Delta z)}{\Delta z} \\ &= \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\frac{\Delta x^3 - \Delta y^3 + i(\Delta x^3 + \Delta y^3)}{\Delta x^2 + \Delta y^2}}{\Delta x + i\Delta y} \\ &= \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\left[\Delta x^3 - \Delta y^3 + i(\Delta x^3 + \Delta y^3) \right] (\Delta x - i\Delta y)}{(\Delta x^2 + \Delta y^2)^2} \\ &= \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y = k \cdot \Delta x}} \frac{\left[\Delta x^3 (1 - k^3) + i\Delta x^3 (1 + k^3) \right] \Delta x (1 - ki)}{\Delta x^4 (1 + k)^2} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\left[1 - k^3 + i(1 + k^3) \right] (1 - ki)}{(1 + k)^2} \end{aligned}$$

$\therefore k$ 不同时，此极限趋于不同值 $\Rightarrow f(z)$ 在 $z=0$ 不可导.

11、证明：如果函数 $f(z) = u + iv$ 在区域 D 内解析，并满足下列条件之一，那么 $f(z)$ 是常数

(3) $\overline{f(z)}$ 在 D 内解析；

解：

$$f(z) \text{ 解析} \Rightarrow \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\overline{f(z)} = u(x, y) - iv(x, y) \text{ 解析} \Rightarrow \frac{\partial u}{\partial x} = -\frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$$

$$\Rightarrow \left. \begin{aligned} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = -\frac{\partial v}{\partial y} &\Rightarrow \frac{\partial v}{\partial y} = 0 = \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = \frac{\partial v}{\partial x} &\Rightarrow \frac{\partial v}{\partial x} = 0 = \frac{\partial u}{\partial y} \end{aligned} \right\} f(z) \text{ 为复常数}$$

12、求下列各式的值

$$(2) e^{\frac{2-\pi i}{3}}$$

$$= e^{\frac{2}{3}} \left[\cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right) \right]$$

$$= e^{\frac{2}{3}} \left(\cos\frac{\pi}{3} - i \sin\frac{\pi}{3} \right)$$

$$= \frac{1}{2} e^{\frac{2}{3}} (1 - i\sqrt{3})$$

13、求 $\text{Ln}(-i), \text{Ln}(-3+4i)$ 和它们的主值

$$\begin{aligned}\text{Ln}(-i) &= \ln|-i| + i\left(-\frac{\pi}{2} + 2k\pi\right) \\ &= i\left(-\frac{\pi}{2} + 2k\pi\right), k = 0, \pm 1, \pm 2, \dots\end{aligned}$$

$$\text{主值: } \ln(-i) = -\frac{\pi}{2}i$$

$$\begin{aligned}\text{Ln}(-3+4i) &= \ln|-3+4i| + i\left(\pi - \arctan\frac{4}{3} + 2k\pi\right) \\ &= \ln 5 + i\left(\pi - \arctan\frac{4}{3} + 2k\pi\right), k = 0, \pm 1, \pm 2, \dots\end{aligned}$$

$$\text{主值: } \ln(-3+4i) = \ln 5 + i\left(\pi - \arctan\frac{4}{3}\right)$$

14、计算下列各式的值

$$\begin{aligned} (1) \quad (1+i)^{1-i} &= e^{(1-i)\operatorname{Ln}(1+i)} = e^{(1-i)\left(\ln\sqrt{2}+i\frac{\pi}{4}+i2k\pi\right)} \\ &= e^{\ln\sqrt{2}+\frac{\pi}{4}+2k\pi+i\left(-\ln\sqrt{2}+\frac{\pi}{4}+2k\pi\right)} \\ &= \sqrt{2}e^{\frac{\pi}{4}+2k\pi}\left[\cos\left(\frac{\pi}{4}-\ln\sqrt{2}\right)+i\sin\left(\frac{\pi}{4}-\ln\sqrt{2}\right)\right] \\ &\quad k=0,\pm1,\pm2,\dots \end{aligned}$$

15、解下列复数方程

$$(2) \ln z = \frac{\pi}{2}i$$

$$\ln z = \ln|z| + i \arg z = \frac{\pi}{2}i$$

$$\therefore |z| = 1, \arg z = \frac{\pi}{2}$$

$$\begin{aligned}\therefore z &= \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \\ &= i\end{aligned}$$

$$(3) 1 + e^z = 0$$

$$e^z = -1$$

$$\begin{aligned}\therefore z &= \operatorname{Ln}(-1) \\ &= \ln 1 + i(\pi + 2k\pi) \\ &= (2k + 1)\pi i\end{aligned}$$

$$k = 0, \pm 1, \pm 2, \dots$$



第三章 作业

北京理工大学

1. 计算积分 $\int_C [(x-y) + ix^2] dz$, 其中C为: (3) 从原点沿虚轴至*i*, 再由*i*沿水平方向向右至1+i.

解: C_1 : 从原点沿虚轴至*i*,

$$C_1 : \begin{cases} x = 0 \\ y = t \end{cases}, t \in [0, 1]$$

$$z = it \rightarrow dz = i dt$$

C_2 : 由*i*沿水平方向向右至1+i

$$C_2 : \begin{cases} x = t \\ y = 1 \end{cases}, t \in [0, 1]$$

$$z = t + i \rightarrow dz = dt$$

$$\begin{aligned} \int_C [(x-y) + ix^2] dz &= \int_0^1 (-t) i dt + \int_0^1 [(t-1) + it^2] dt \\ &= -i \cdot \frac{1}{2} t^2 \Big|_0^1 + \frac{1}{2} (t-1)^2 \Big|_0^1 + \frac{i}{3} t^3 \Big|_0^1 = -\frac{1}{2} - \frac{1}{6} i \end{aligned}$$

3.计算积分 $\int_C |z| dz$, 其中C为:

(3) 从-1沿下半单位圆周至1.

解:
$$\begin{cases} x = \cos t \\ y = \sin t \end{cases}, -\pi \leq t \leq 0$$

$$z = \cos t + i \sin t, dz = (-\sin t + i \cos t) dt$$

$$\begin{aligned} \int_C |z| dz &= \int_{-\pi}^0 (-\sin t + i \cos t) dt \\ &= \cos t \Big|_{-\pi}^0 + i \sin t \Big|_{-\pi}^0 \\ &= 2 \end{aligned}$$

4. 设C为从原点到 $4+3i$ 的直线段, 求 $\left| \int_C \frac{1}{z-i} dz \right|$ 的最大值。

解: $C : z = (4+3i)t, t \in [0,1]$

$$\left| \frac{1}{z-i} \right| = \left| \frac{1}{(4+3i)t-i} \right| = \frac{1}{\sqrt{(5t-\frac{3}{5})^2 + \frac{16}{25}}} \leq \frac{5}{4}$$

根据积分的估值不等式有

$$\begin{aligned} \left| \int_C \frac{1}{z-i} dz \right| &\leq \int_C \left| \frac{1}{z-i} \right| dl \leq \frac{5}{4} \int_C dl \\ &= \frac{5}{4} \times \sqrt{3^2 + 4^2} = \frac{25}{4} \end{aligned}$$

5.试用观察法求得下列积分的值，并说明依据， C 是正向单位圆周 $|z|=1$.

$$(3) \oint_C \frac{dz}{\cos z} \qquad (5) \oint_C \frac{dz}{(z - \frac{1}{2})^5}$$

(3) 令 $\cos z = 0 \Rightarrow$ 奇点 $z = k\pi + \frac{\pi}{2}, k = 0, \pm 1, \pm 2, \dots$

奇点最小模为 $\frac{\pi}{2} > 1 \Rightarrow \frac{1}{\cos z}$ 在 C 上及内部处处解析，

$\therefore$ 根据柯西-古萨定理，原式 = 0.

$$(5) \oint_C \frac{dz}{(z - \frac{1}{2})^5} = \frac{2\pi i}{4!} 1^{(4)} = 0$$

7.计算下列积分:

$$(3) \oint_C \frac{1}{(z^2-1)(z^3-1)} dz, C: |z|=r < 1;$$

$$(6) \oint_C \frac{z^2-3z+4}{z(z-2)^2} dz, \text{其中 } C: |z|=1 \text{ 为正方向}$$

解: (3) 由 $z^2=1, z^3=1 \Rightarrow$ 奇点都在 $|z|=1$ 上,
即在 C 内部均解析 $\Rightarrow$ 原积分 $= 0$

$$(6) \text{原式} = \oint_C \frac{\frac{z^2-3z+4}{(z-2)^2}}{z} dz = 2\pi i \left[\frac{z^2-3z+4}{(z-2)^2} \right] \Big|_{z=0} = 2\pi i$$

8. 计算 $\oint_C \frac{1}{z(z^2+1)} dz$, 其中C为: (2) $|z|=2$;

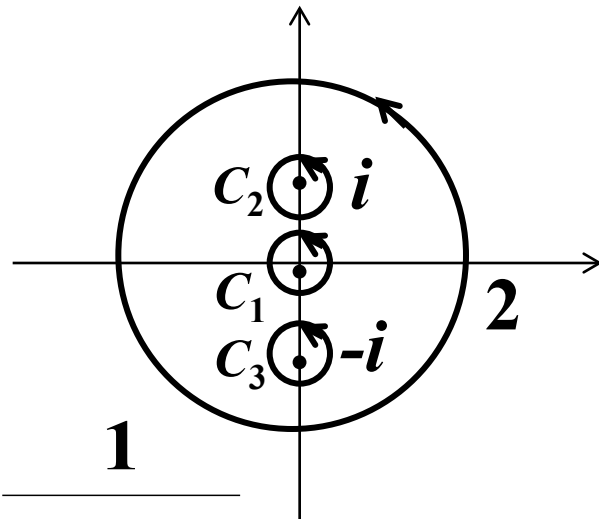
解:

$$\oint_C \frac{1}{z(z^2+1)} dz$$

$$= \oint_{C_1} \frac{1}{z} dz + \oint_{C_2} \frac{1}{z-i} dz + \oint_{C_3} \frac{1}{z+i} dz$$

$$= 2\pi i \left(\frac{1}{z^2+1} \right) \Big|_{z=0} + 2\pi i \left(\frac{1}{z(z+i)} \right) \Big|_{z=i} + 2\pi i \left(\frac{1}{z(z-i)} \right) \Big|_{z=-i}$$

$$= 2\pi i + 2\pi i \left(-\frac{1}{2} \right) + 2\pi i \left(-\frac{1}{2} \right) = 0$$



9.计算下列积分:

$$(3) \int_{-\pi i}^{\pi i} \sin^2 z dz;$$

解:
$$\begin{aligned} \int_{-\pi i}^{\pi i} \sin^2 z dz &= \int_{-\pi i}^{\pi i} \left(\frac{1 - \cos 2z}{2} \right) dz \\ &= \frac{1}{2} \left(2\pi i - \frac{1}{2} \sin 2z \Big|_{-\pi i}^{\pi i} \right) \\ &= \frac{1}{2} (2\pi i - \sin 2\pi i) = \pi i - \frac{1}{2} \sin 2\pi i \\ &= \pi i - \frac{1}{2} \cdot \frac{e^{i \cdot 2\pi i} - e^{-i \cdot 2\pi i}}{2i} = \pi i + \frac{1}{4} i (e^{-2\pi} - e^{2\pi}) \end{aligned}$$

11. 计算下列积分:

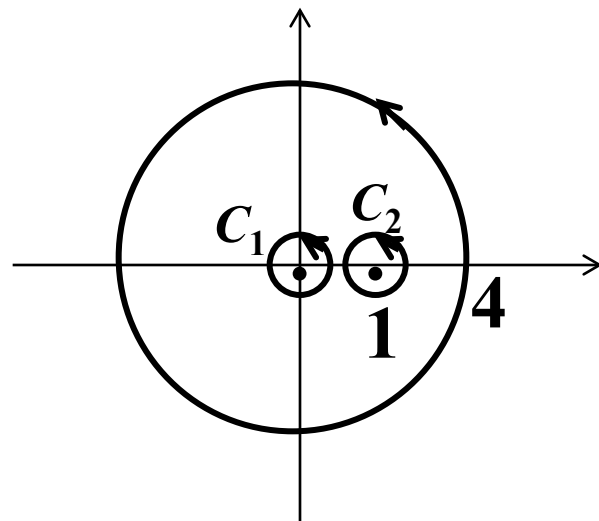
$$(2) \oint_C \frac{e^z dz}{z^2 (z-1)^2}, C: |z|=4;$$

$$= \oint_{c_1} \frac{e^z}{z^2} dz + \oint_{c_2} \frac{e^z}{(z-1)^2} dz$$

$$= 2\pi i \left[\frac{e^z}{(z-1)^2} \right]' \bigg|_{z=0} + 2\pi i \left[\frac{e^z}{z^2} \right]' \bigg|_{z=1}$$

$$= 2\pi i \left[\frac{e^z (z-1)^2 - e^z 2(z-1)}{(z-1)^4} \right] \bigg|_{z=0} + 2\pi i \left[\frac{e^z z^2 - e^z 2z}{z^4} \right] \bigg|_{z=1}$$

$$= 2\pi i(1+2) + 2\pi i(e-2e) = 2\pi i(3-e)$$



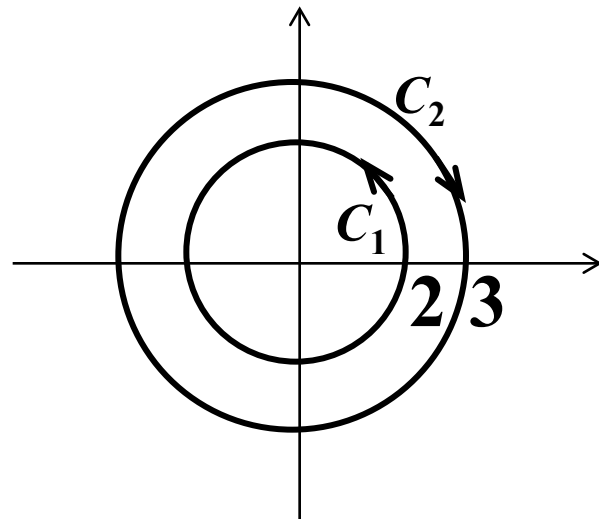
(4) $\oint_{C=C_1+C_2} \frac{\cos z}{z^3} dz$, 其中 $C_1 : |z|=2$ 为正方向,
 $|z|=3$ 为负方向;

解:

$\therefore \frac{\cos z}{z^3}$ 在复合闭路 C^- 内解析,

$$\therefore \oint_{C^-} \frac{\cos z}{z^3} dz = 0$$

$$\therefore \oint_C \frac{\cos z}{z^3} dz = -\oint_{C^-} \frac{\cos z}{z^3} dz = 0$$



14.由下列条件求解析函数 $f(z) = u + iv$.

(3) $v = \frac{y}{x^2 + y^2}, f(2) = 0.$

解: $\frac{\partial v}{\partial x} = \frac{-2xy}{(x^2 + y^2)^2}, \frac{\partial v}{\partial y} = \frac{x^2 + y^2 - 2yy}{(x^2 + y^2)^2} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$

$$f'(z) = \frac{\partial v}{\partial y} + i \frac{\partial v}{\partial x} = \frac{x^2 - y^2}{(x^2 + y^2)^2} + i \frac{-2xy}{(x^2 + y^2)^2}$$

令 $x = z, y = 0$, 则 $f'(z) = \frac{z^2}{z^4} = \frac{1}{z^2} \Rightarrow f(z) = -z^{-1} + c$
 C 为实常数

$$f(2) = 0 = -\frac{1}{2} + c \Rightarrow c = \frac{1}{2} \Rightarrow f(z) = -\frac{1}{z} + \frac{1}{2}$$

16. 设 $f(z) = u + iv$ 为解析函数,

证明: $-u$ 是 v 的共轭调和函数。

证明1:

$f(z)$ 解析 $\Rightarrow u$ 和 v 存在一阶连续偏导且 $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

$\therefore v$ 和 $-u$ 存在一阶连续偏导

且 $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = \frac{\partial(-u)}{\partial y}, \frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = -\frac{\partial(-u)}{\partial x}$ 即满足 $C-R$ 方程,

$\therefore g(z) = v + (-u)i$ 为解析函数 $\Rightarrow -u$ 是 v 的共轭调和函数.

证明2: $\because f(x)$ 为解析函数,

$\therefore g(z) = -i(u + iv) = -if(x)$ 为解析函数,

$\therefore g(z) = v + (-u)i$ 为解析函数 $\Rightarrow -u$ 是 v 的共轭调和函数.